EDC · T0–1

Circuit tools and semiconductor physics

Cheat sheet · Day 01 · Sat 24 Oct · pack E1

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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E1 - CIRCUIT TOOLS + SEMICONDUCTOR PHYSICS
Divider (series only): Vi = Vin*Ri/Rsum   Current div (2 R): I1 = I*R2/(R1+R2)
Thevenin: Voc with load removed; RTh = R seen with V->short, I->open
  dep. sources present: RTh = Voc/Isc ; only dep.: RTh = Vtest/Itest
  IN = VTh/RTh, RN = RTh
Bias node X fed by (Vk,Rk), ground = 0 V branch:
  VTh = sum(Vk/Rk)/sum(1/Rk)    RTh = R1||R2||...
  Divider: VTh = VCC*R2/(R1+R2), RTh = R1||R2  -> then base KVL:
  VTh = IB*RTh + VBE + IE*RE
Nodal: ref node, KCL (leaving = source in), V-src between nodes -> supernode
Superposition: one indep. source at a time; dep. sources stay ON
Ideal op-amp: v+ = v-, i+ = i- = 0, no KCL at output; Vo limited by rails
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ni = B T^1.5 exp(-Eg/2kT)  B=7.3e15, Eg=1.12 eV, k=8.62e-5 eV/K, T in K
ni(300K) = 1.5e10 cm^-3 ; ni^2 = 2.25e20
n0*p0 = ni^2 (equilibrium)
n-type: n ~ ND, p = ni^2/ND ; p-type: p ~ NA, n = ni^2/NA
Exact (n-type): n = ND/2 + sqrt((ND/2)^2 + ni^2), p = ni^2/n  (if ni >~ 0.1 ND)
sigma = q(n mu_n + p mu_p) (ohm-cm)^-1 ; rho = 1/sigma ; R = rho L/A (cm!)
vd = mu E, E = V/L ; J = sigma E ; D/mu = VT (Einstein)
mu: WS uses 1350/500, lecture slide 1300/500 -> use what is given
VT = kT/q, k = 1.38e-23 J/K, q = 1.6e-19 ; 27C -> 25.875 mV
1 um = 1e-4 cm ; 1 um^2 = 1e-8 cm^2 ; 1 mm = 0.1 cm
pn junction: open ckt ID (diffusion, majority) = IS (drift, minority)
  xn/xp = NA/ND (extends into lightly doped side), W ~ 0.1-1 um
  V0 = VT ln(NA ND / ni^2) ; FB: W shrinks, I exp ; RB: W grows, only IS
  IS doubles per 10 C ; doped material is electrically NEUTRAL
Group V donor -> n-type ; Group III acceptor -> p-type
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